ASTERION

DESK MANUAL

A beginner's guide to every feature, in plain words

No crypto background assumed. If you can open Telegram and type, you can run this.



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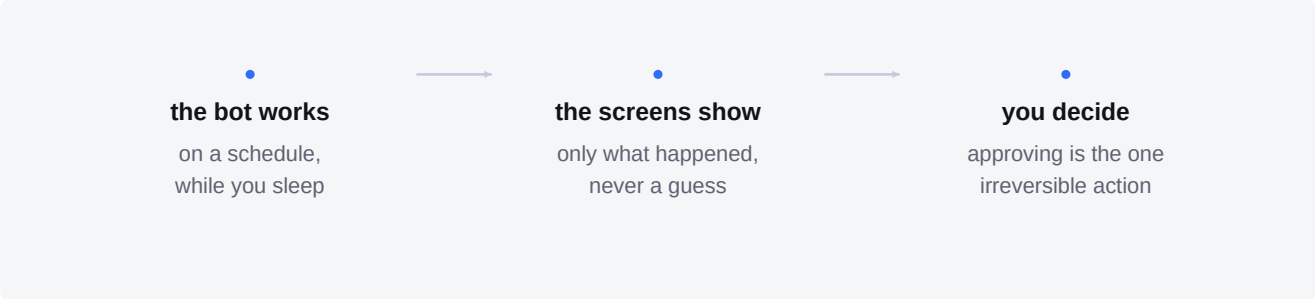
What this thing is

Asterion is two workers in one program. One **hunts for bugs in crypto software** and pays for itself by finding holes before someone else does. The other **trades on paper** - fake money, real prices - to find out whether a strategy works before you risk a single real dollar. Both run on a schedule, on a server, while you sleep.

You read two screens: one for the desk, one for the trading. That is the whole system.

1 The safest part is the part you cannot rush

Nothing here can spend your money until seven separate conditions are true. That is not caution for its own sake - it is the design.



- **Nothing here is verified against a live exchange yet.** The desk runs on paper. Its own arming gate refuses until seven conditions hold, and even then live order submission is not implemented. You cannot lose money by using this today.
- **Silence is the normal state.** An empty queue and an empty trade list are correct most days. If you need daily reassurance that something is happening, this will feel broken - and that feeling is the feature working.

2

The five words you need

Everything else in this document is detail. These five are the vocabulary the rest of it uses.

Paper mode

Trading with fake money. Prices are real, orders are real, the money is imaginary. You lose nothing and learn everything.

Live mode

Trading with real money. Same code, real orders, real risk. Seven gates stand between you and this.

A position

Something you bought and still own. Bought $50 of a coin and hold it? That is an open position of $50.

A stop

An automatic sell that fires when the price falls to a level set in advance. It is what stops one bad day becoming a bad month. The desk sets it for you.

Slippage

The gap between the price you expected and the price you got. On small coins, your own sell moves the price down against you. It happens on the way in and the way out, and the desk models both.

3

What actually runs, and when

Nothing in this system is a button you press repeatedly. It is a schedule. Five timers run on the server, and you do not have to be awake, online, or have your laptop open.

every 5 min	**Sentinel + desk tick**	scans the watchlist, then one full trading pass
every hour	**Crawler + analyst**	fetches new bounty programs, re-scores what passed
03:15 daily	**Maintenance**	verifies the audit log, ages out old approvals
every 6 hours	**Dead-man's switch**	checks the worker is alive, alerts if not
06:30 daily	**Briefing**	one Telegram message with the day's summary

3 The 5-minute tick is the important one

It does four things in a fixed order, and the order matters - each stage is cheaper than the next, so the expensive ones only see survivors.

1 SCOUT

liquid?
new enough?

throws away the obviously wrong

→

2 FILTER

can you sell?
owner mint?

simulates a real sell, checks the mint authority

→

3 ANALYST

score 0-100
any floor failed?

writes the number and the reason

→

4 RISK

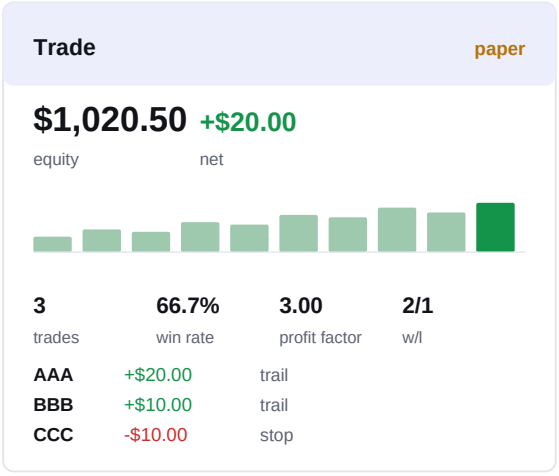
slot free?
daily loss hit?

opens at half size, sets the stop

ANY STAGE CAN KILL A COIN most do, most days, and that is the feature working - an empty trade list is a correct outcome, not a broken bot

The Trade screen

This is the screen that looks like a trading app. Read top to bottom, and read the blocks in the order below - each one answers a narrower question than the one above it.



4 The one number people misread

Profit factor is money won divided by money lost. It is not a score out of anything.

- **Above 1.0** - profitable. **Above 1.3** - good. **Above 2.0** - suspicious, and assume it will not last.
- If it reads **n/a (no losses yet)**, that is arithmetic, not a great sign. There is nothing to divide by yet.
- Under 30 closed trades, ignore every number on the screen. It is noise dressed as a result.

5

Hunt targets

This screen answers a different question: **what is the desk looking at right now, and what did it already throw away?** It reads the table the desk actually fills.

THE FIVE STATES, IN PIPELINE ORDER

passed	survived every gate and is tradeable right now. What the desk will buy from.
scored	survived the cheap checks and got a score, but no position opened - usually no free slot.
in (raw)	the scout saw it and has not checked it properly yet. A queue, not a candidate.
killed	rejected, with a reason. The most informative number on the screen: a desk that kills nothing is not filtering.
expired	too old to be interesting. Rows age out.

5 Read the why line, not the score

A token scoring 78 that was still killed has a why line naming the single floor it missed. That reason is far more useful than the total.

- **Liquidity** is how much money sits in the pool you would trade against. A $40k pool against a $20M market cap means your own sell moves the price, and the desk therefore requires liquidity of at least **3% of market cap**.
- **Holders** is how many separate wallets own it. Ten holders means one wallet selling crashes it.

6

The Library, and what changed

The Library lists **what the program can actually do**. Each entry is one bug class it knows how to hunt, and each shows two facets that are deliberately separate.

a playbook

the exact route and proof method. If it has one, the bot can actually run it against a target and prove a result.

a method pack

the questions to ask first, the false positives to rule out, and what a proof must show. The thinking, not the executing.

6 What changed this round

The Solidity set went from 13 routed classes to 17, and 20 packs in total. Every one of the seven is new methodology, a new route, and a new proof harness.

sol.overflow	arithmetic that wraps, or a cast that silently truncates
sol.mev	sandwich, backrun and ordering attacks
sol.erc4626	vault share price and the first-depositor donation
sol.erc20	tokens that take a fee, rebase, or return nothing
sol.proxy	storage slots that collide between proxy and logic
sol.erc4337	account abstraction and paymaster drains
sol.signature.ecdsa	signature replay and zero-address recovery

Also fixed: Solana was filed under a heading that read "Solidity"

The three Solana classes have ids starting with *sol.* and were grouped by that prefix. They are Solana programs. The class now decides the heading, and there is a Solana group.

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Reading the rest of the screens

Four screens remain, and each has exactly one job.

Hunt

Describe a target in plain text. The bot compiles it into a real engagement with rules, and puts it in the queue for you to approve. Nothing runs until you approve.

Queue

The approvals waiting for you. Approving is the one irreversible action in the console, so it asks first and writes to an append-only log that cannot be edited afterwards.

Ask

Talk to the bot in sentences. It answers about the system.

Ops

The inventory: programs, findings, history, the audit log, the desk, the Trade screen, the Hunt-target screen, status and the scoreboard.

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How a trade actually happens

The full path, once, end to end.

1. The scout sees a new coin with enough liquidity and volume, and writes a row.
2. The filter checks the dangerous things: can the owner mint unlimited tokens, can they freeze your wallet, is selling actually possible.
3. The analyst scores it. Failing the floor on any single check kills it, regardless of the total.
4. Risk checks for a free position slot and whether the day's loss limit is already hit. If both are fine, it opens at **half size**.
5. The other half is bought a moment later at a confirmed price, which is what stops a fast-moving price filling you at a bad level.
6. A stop is set immediately, and it only ever moves up.
7. When the stop, the trail, or the thesis fires, it sells and writes the exit with a reason code.
8.

Two hard limits, and why they are the whole safety

Both are floor limits on the damage, not targets for the upside.

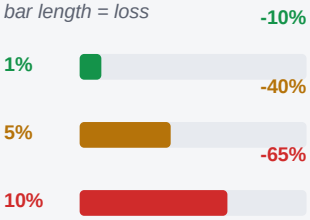
10 losses in a row, at three risk sizes

compounded, not added

risk / trade	left after 10	still needed to recover
1%	$9,044 of $10,000	+11% to get back
5%	$5,987 of $10,000	+67% to get back
10%	$3,487 of $10,000	+187% to get back

how deep each hole is

bar length = loss



The 1% column is survivable. The 10% column needs a double just to break even. Same strategy, same ten trades.

9

Going live, and why it takes seven checks

Paper mode is not live mode with a number changed. Each gate below exists because something went wrong once without it.

ALL SEVEN, OR IT STAYS ON PAPER

no button flips this - there is no button that can start real trading

1	**200 closed paper trades**	fewer than that and the statistics mean nothing
2	**30 days of paper history**	a strategy can look great for a week and be broken by a month
3	**positive expectancy after fees**	it must make money after costs, not before
4	**max drawdown under 20%**	the worst peak-to-trough run has to be survivable
5	**DESK_LIVE=1 set as a secret**	a deliberate act, never a default
6	**a capital cap above zero**	a hard ceiling on what can be lost
7	**the signing key held outside the worker**	the key that moves money must not sit with the code that decides

9 There is deliberately no go-live button

Arming is a server-side secret, not a web control. A web control that can start real trading is one leaked URL away from an empty wallet.

Until all seven are true the mode stays **paper** no matter what you type. You can ask **/desk live** in Telegram and it will list what is missing. It cannot arm itself.

The Telegram commands

Everything the web console does, the bot does too, in the chat.

/start /menu	the main menu
/status	worker alive, how many programs, what is queued
/hunt <target>	start a hunt on something
/desk	desk status and the unmet gates
/pos	open positions
/kills	what was rejected and why
/calib	how accurate the filters have been
/corpus	the precedent data behind the scoring
/ui	a link to the full console

10 The honest state of the desk

Two things are true right now, and both matter more than any feature list.

- **The flow gate that used to block judging is closed.** The filter refuses to judge a token whose flow fields are null, so every candidate was written as a coverage gap and nothing ever opened. The provider that closes it is wired in, so the desk reaches a verdict now.
- **Live order submission is still unimplemented.** The executor throws *live_signer_not_wired*, so even a fully armed desk refuses. It is a paper-only research instrument until a signer is wired. That is the part of the README that remains true.

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What empty means, in every place it appears

Re-read this section whenever something looks blank. Empty is usually correct, and each empty has a different cause.

- **No closed trades**

  Paper mode has not exited a position yet. Positions only close on a stop, a trail, or a thesis break.

- **No open positions**

  Every slot was empty and nothing passed the gates. Normal, and the point.

- **No targets**

  The scout has not written a row since deploy. The tick runs every 5 minutes; wait one. If still zero, every candidate failed a coarse gate before a row existed.

- **No approvals**

  Nothing is waiting on you. Correct.

- **No findings**

  The bot has not proven a bug yet. A finding is written when proven, not when suspected.

- **Method only**

  That Library class can be reasoned about but has no automated proof route. Honest, and labelled as such.

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The five things that will cost you money

If you let them. Each one has a defence already built in.

1. Overfitting

A strategy tuned on past data fits that data and nothing else. That is why the desk holds back part of the history the strategy never saw, and why going live needs 30 days rather than 30 trades.

2. Thin liquidity

A $40k pool cannot absorb your exit. You will see a paper profit that cannot be realised. The 3%-of-market-cap rule is the defence.

3. One bad week

Which is why the daily breaker and the 1% risk exist. Ten 1% losers is 10%, survivable. Ten 10% losers is ruin.

4. A honeypot

A token that lets you buy and blocks your sell. The filter checks it by simulating a sell, because reading the code is not enough.

5. Believing the backtest

Every backtest is a story about the past. The only number that has ever predicted the future is live results, and even those only after a few hundred trades.

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The one-minute version

THE DAILY LOOP

- Open the console.
- Look at Trade: read equity and net, then the shape of the curve.
- Look at Hunt targets: check the passed count, read the why lines on anything killed.
- If passed is zero and the kills have real reasons, everything is working.
- Leave it alone. Check again tomorrow.

The desk is not a slot machine. It is a filter that occasionally finds something worth doing. Most days it will do nothing, and that is the feature.

Asterion · desk manual

Generated from the same source the screens read. Numbers in the figures are computed, not typed.